

DR MATTHIAS ALLARD

PhD (Mathematics) | MSc (Theoretical Physics) | Diplôme d'Ingénieur (Photonics, MSc level)

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RESEARCH INTERESTS

I work at the interface of probability theory, random matrix theory, and mathematical physics. My research focuses on universal phenomena in random matrix ensembles, including eigenvalue and singular value correlations, hard-edge asymptotics, and determinantal structures. I am particularly interested in the connections between random matrices, Coulomb gases, and integrable probability, developing analytical approaches based on complex analysis, orthogonal polynomials, and asymptotic methods.

TECHNICAL SKILLS

- **Programming Languages** : Python, Mathematica, C++, MATLAB, LaTeX
- **Scientific Computing** : Numerical Simulation, Mathematical Modeling, Asymptotic Analysis, Scientific Computing, Image Processing
- **Scientific Areas** : Random Matrix Theory, Probability, Complex Analysis, Statistical Physics, Optical Engineering, Signal Processing
- **Languages** : French (native), English (fluent), Spanish (intermediate)

EDUCATION

- **Joint PhD in Random Matrices and Complex Analysis** 2021 – 2025
University of Melbourne & KU Leuven *Australia / Belgium*
 - Joint PhD under the supervision of Mario Kieburg (University of Melbourne) and Arno Kuijlaars (KU Leuven)
 - Research topics: Random Matrix Theory, Probability, Harmonic Analysis, Orthogonal Polynomials, Measure Theory, Coulomb Gases
 - Recipient of the Melbourne Research Scholarship
- **MSc Physics (Double Diploma), with Distinction** 2019 – 2020
Imperial College London *London, UK*
 - Quantum Information, Quantum Field Theory, Unification, General Relativity
 - Mathematical Methods for Physicists, Plasmonics and Metamaterials, Group Theory
- **Diplôme d'Ingénieur Photonics & MSc Fundamental Physics, (concurrent)** 2018 – 2019
Institut d'Optique Graduate School & Université Paris-Saclay *Paris, France*
 - **IOGS (Engineering / Photonics)**: Fourier Optics, Nonlinear Optics, Radiometry, Optical System Design, Light–Matter Interaction
 - **Université Paris Saclay (Physics)**: Atomic Physics, Plasma Physics, Particle Physics
- **Diplôme d'Ingénieur Photonics & BSc Fundamental Physics, (concurrent)** 2017 – 2018
Institut d'Optique Graduate School & Université Paris-Saclay *Paris, France*
 - **IOGS (Engineering / Photonics)**: Signal Processing, Polarization, Laser Physics, Scientific Computing, Optical Physics
 - **Université Paris Saclay (Physics)**: Quantum Physics, Statistical Physics, Special Relativity, Analytical Mechanics, Electromagnetism
- **Classe Préparatoire aux Grandes Écoles** 2015 – 2017
Lycée Chaptal *Paris, France*
 - Two-year intensive undergraduate preparation in Mathematics (14h/week), Physics (12h/week), and Engineering (12h/week)
- **Baccalauréat S (Science), Engineering Option, with Distinction** 2015
France

RESEARCH AND ENGINEERING EXPERIENCE

- **Research Project** 2020
Imperial College London London, UK
– Studied Directed Acyclic Graphs, Minkowski distances, and percolation models
- **Research Engineering Intern** 2019
Thales Alenia Space Cannes, France
– Modeled optical fiber components within an imaging spectrometer
– Conducted calibration and modeling of an imaging Fourier-transform spectrometer
– Studied speckle phenomena and optical signal processing
- **Engineering Project** 2018 – 2019
Institut d'Optique Graduate School France
– Designed and implemented an industrial vision system
– Worked on programming, mechanical engineering, and electrical engineering components
- **Research Intern** 2018
Observatoire des Hauts de Provence France
– Worked on image processing, speckle analysis, infrared detectors, and astronomical observations
- **Sports Activities Coordinator** 2018 – 2019
Institut d'Optique Graduate School France
– Organized events, coordinated teams, and planned student activities

SELECTED PUBLICATIONS

- Allard, M., Kieburg, M. *Correlation functions between singular values and eigenvalues*. Random Matrices: Theory and Applications, 15 (2025), 2550024
- Allard, M. *Hard edge asymptotics of correlation functions between singular values and eigenvalues*. Journal of Physics A: Mathematical and Theoretical, 58 (2025), 215204
- Allard, M., Lahiry, S. *Critical phenomena in Coulomb gas: birth of a gap*. arXiv:2509.24529. To appear in Communications in Mathematical Physics.
- Allard, M., Forrester, P., Lahiry, S., Shen, B. *Partition function of 2D Coulomb gases with radially symmetric potentials and a hard wall*. arXiv:2506.14738

TEACHING EXPERIENCE

- **Tutor** 2023 – 2026
University of Melbourne Australia
– Taught Calculus 1, Calculus 2, Linear Algebra, and Physics laboratories
- **Tutor** 2021 – 2022
KU Leuven Belgium
– Taught Calculus 1 and Differential Equations

SCIENTIFIC CONFERENCES

- 2026
 - Gathering Advances in Random Matrices and Beyond (IHP, France)
- 2025
 - ANZAMP 2025 (Bendigo)
 - Integrability, Random Matrices, and All That (ENS Lyon)
 - Log-gases in Caeli Australi (MATRIX, Creswick)
 - Randomness in Physics and Mathematics Summer School (Bielefeld)
- 2024
 - Asymptotics, Randomness, Nonlinearity, Orthogonality (ARNO, KU Leuven)
 - Random Matrices and Related Topics in Jeju (Jeju)
- 2022
 - Random Matrices and Beyond (KTH, Stockholm)
 - Probability and Mathematical Physics (University of Helsinki, Helsinki)
 - Randomness in Physics and Mathematics Summer School (Bielefeld)